

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****ASSESSMENT TOOL 2**

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4	Assessment:	Assessment Tool 2
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:	CSE-AI	Date:	24/08/2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ G.VEDAVIKAS GOWD _____

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Industry Problem Solving

Question 1 – Digital Marketing

Question:

Before introducing a new advertisement, the conversion rate was 8%. After the campaign, 90 out of 1,000 visitors converted. Using a 95% confidence interval, determine whether the observed conversion rate provides evidence of improvement.

Answer

Given:

Previous conversion rate:

$$[p_0=0.08]$$

Sample size:

$$[n=1000]$$

Number of conversions:

$$[x=90]$$

Sample conversion rate:

$$[\hat{p}=\frac{90}{1000}=0.09]$$

For 95% confidence level:

$$[Z_{\alpha/2}=1.96]$$

Standard Error:

$$[SE=\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}]$$

$$[SE=\sqrt{\frac{0.09(0.91)}{1000}}]$$

$$[SE=0.00905]$$

95% Confidence Interval:

$$[CI=\hat{p}\pm 1.96(SE)]$$

$$[=0.09 \pm 1.96(0.00905)]$$

$$[=0.09 \pm 0.01774]$$

$$[CI=(0.07226,;0.10774)]$$

Therefore,

$$[CI=(7.23\%,10.77\%)]$$

Conclusion:

The original conversion rate of 8% lies within the 95% confidence interval. Therefore, the increase from 8% to 9% is **not statistically significant**. There is **insufficient evidence to conclude that the new advertisement improved the conversion rate**.

Question 2 – Education Technology

Question:

An EdTech company claims that its adaptive-learning platform improves the average test score to at least 75. A sample of 64 students obtains an average score of 78 with standard deviation 10. Use a 95% confidence interval to assess the claim.

Answer

Given:

$$[n=64]$$

$$[\bar{x}=78]$$

$$[s=10]$$

Confidence level = 95%

Degrees of freedom:

$$[df=64-1=63]$$

For 95% confidence:

$$[t_{\{0.025,63\}} \approx 2.00]$$

Standard Error:

$$[SE = \frac{s}{\sqrt{n}}]$$

$$[SE = \frac{10}{\sqrt{64}}]$$

$$[SE = \frac{10}{8} = 1.25]$$

Margin of Error:

$$[ME = t \times SE]$$

$$[ME = 2.00(1.25) = 2.50]$$

95% Confidence Interval:

$$[CI = \bar{x} \pm ME]$$

$[CI=78 \pm 2.50]$

$[CI=(75.50, 80.50)]$

Conclusion:

The 95% confidence interval for the average test score is **(75.50, 80.50)**. Since the entire interval is above 75, there is **sufficient evidence to support the company's claim** that the average test score is at least 75.

Question 3 – Automobile Industry

Question:

A car manufacturer claims that its new model provides an average mileage of 20 km/litre. A sample of 50 cars gives a mean mileage of 19.2 km/litre with standard deviation 2.5 km/litre. Perform a hypothesis test at the 5% significance level using the p-value approach.

Answer

Given:

$$[\mu_0=20]$$

$$[n=50]$$

$$[\bar{x}=19.2]$$

$$[s=2.5]$$

$$[\alpha=0.05]$$

Step 1: Hypotheses

$$[H_0:\mu=20]$$

$$[H_1:\mu\neq 20]$$

Step 2: Test Statistic

Since the population standard deviation is unknown, use the t-test.

$$[t=\frac{\bar{x}-\mu_0}{s/\sqrt{n}}]$$

$$[t=\frac{19.2-20}{2.5/\sqrt{50}}]$$

$$[t=\frac{-0.8}{0.3536}]$$

$$[t=-2.26]$$

Degrees of freedom:

$$[df=50-1=49]$$

Step 3: p-value

For ($t=-2.26$) with 49 degrees of freedom:

$$[p\text{-value}] \approx 0.028$$

Step 4: Decision

$$[p\text{-value}] < \alpha$$

$$[0.028 < 0.05]$$

Therefore, **reject (H_0)**.

Conclusion

There is sufficient evidence at the 5% significance level to conclude that the average mileage is significantly different from 20 km/litre. Since the sample mean is 19.2 km/litre, the actual mileage appears to be **lower than the claimed 20 km/litre**.

Question 4 – Cloud Computing

Question:

A cloud provider claims that its average server response time is 100 ms. A sample of 81 requests has a mean of 104 ms and standard deviation of 12 ms. Test the claim at ($\alpha=0.05$) using the p-value method.

Answer

Given:

$$[\mu_0=100]$$

$$[n=81]$$

$$[\bar{x}=104]$$

$$[s=12]$$

$$[\alpha=0.05]$$

Step 1: Hypotheses

$$[H_0:\mu=100]$$

$$[H_1:\mu\neq 100]$$

Step 2: Test Statistic

$$[t=\frac{\bar{x}-\mu_0}{s/\sqrt{n}}]$$

$$[t=\frac{104-100}{12/\sqrt{81}}]$$

$$[t=\frac{4}{12/9}]$$

$$[t=3.00]$$

Degrees of freedom:

$$[df=81-1=80]$$

Step 3: p-value

For ($t=3.00$) with 80 degrees of freedom:

$p\text{-value} \approx 0.0036$

Step 4: Decision

$0.0036 < 0.05$

Therefore, **reject (H_0)**.

Conclusion

There is sufficient statistical evidence at the 5% significance level to conclude that the average server response time is significantly different from 100 ms. Since the sample mean is 104 ms, the actual response time appears to be **higher than 100 ms**.

Question 5 – E-Commerce

Question:

An online company claims that its product return rate is 10%. In a sample of 1,000 orders, 130 products were returned. Test whether the actual return rate is significantly higher than 10% using the p-value approach at the 5% significance level.

Answer

Given:

$$[p_0=0.10]$$

$$[n=1000]$$

Returned products:

$$[x=130]$$

Sample proportion:

$$[\hat{p}=\frac{130}{1000}=0.13]$$

$$[\alpha=0.05]$$

Step 1: Hypotheses

Since we are testing whether the return rate is **higher than 10%**:

$$[H_0:p=0.10]$$

$$[H_1:p>0.10]$$

Step 2: Standard Error

$$[SE=\sqrt{\frac{p_0(1-p_0)}{n}}]$$

$$[SE=\sqrt{\frac{0.10(0.90)}{1000}}]$$

$$[SE=0.00949]$$

Step 3: Test Statistic

$$[z=\frac{\hat{p}-p_0}{SE}]$$

$$[z=\frac{0.13-0.10}{\sqrt{0.00949}}]$$

$$[z=3.16]$$

Step 4: p-value

For ($z=3.16$):

$$[p\text{-value}\approx 0.0008]$$

Step 5: Decision

$$[0.0008 < 0.05]$$

Therefore, **reject (H_0)**.

Conclusion

There is sufficient evidence at the 5% significance level to conclude that the actual product return rate is **significantly higher than 10%**. The sample shows a return rate of 13%, which is significantly greater than the company's claimed rate of 10%.